HACKATHON PROJECT • ROUND 1

# AI-Powered Personal Health Copilot

Understand • Organize • Manage

Team Name: NOVER CODER

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**TEAM MEMBERS**

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# 1. The Problem

Medical information is often scattered across documents and difficult to interpret.

### Scattered records

Prescriptions, lab reports, discharge summaries and diagnostic results are stored in different places.

### Complex medical language

People may struggle to understand test values, abbreviations and unfamiliar medical terms.

### Hard to track history

Comparing reports over time can be confusing when dates and results are not organized together.

## Our goal

Create one simple place where users can upload health records, read plain-language explanations, and review their health-document timeline.

# 2. Our Proposed Solution

A web-based personal health assistant focused on clarity and organization.

01

### Upload

PDF or image of a medical report

02

### Extract

Read text and available report details

03

### Explain

Generate a plain-language summary

04

### Organize

Save records in a date-wise timeline

**The Copilot supports understanding and organization; it does not replace a qualified healthcare professional.**

# 3. Three Core Features

The core workflow required by the challenge.

### 01 Record Intelligence & OCR

- Upload prescriptions and lab reports
- Extract readable text from PDF/ images
- Capture available dates, test names, values and medicine details
- Flag unclear text instead of guessing

### 02 Simple AI Health Summary

- Explain report content in plain language
- Show out-of-range values when reference ranges are available
- Keep explanations factual and understandable
- Avoid diagnosis or medication changes

### 03 Unified Health Profile

- Keep records together in one place
- Display report dates and extracted details
- View history in a timeline
- Help users find previous documents more easily

**Prototype target: a working upload → extraction → summary → timeline flow.**

# 4. Bonus Features

Planned extensions to improve accessibility and technical readiness.

## Bonus A • Multi-language Support

- English and Telugu summary option
- Make medical wording easier to understand
- OCR support for clear printed text first
- Handwriting support only where the OCR engine can read it reliably

## Bonus B • FHIR-Style / Mock ABHA

- Structure demo records using FHIR-inspired JSON
- Example resource types: Patient, Observation, DiagnosticReport
- Show a clearly labelled dummy ABHA-style ID
- No real ABHA account connection in the prototype

These are prototype/demo capabilities; actual integration requires separate validation and authorization.

# 5. Proposed Technical Architecture

A beginner-friendly architecture that can be expanded later.




### Language Layer

English / Telugu display

### Structured Export

FHIR-inspired JSON demo

### Privacy & Safety

Sample data, no diagnosis

**Suggested stack: Python • Streamlit • PDF text extraction • OCR library • JSON/local storage**

# 6. Safety, Privacy & Responsible AI

Healthcare information must be handled carefully.

### Use synthetic demo data

Avoid uploading real patient records during the public demo.

### Do not guess

If OCR cannot read handwriting or a value, mark it as unclear.

### No medical diagnosis

Provide educational explanations, not diagnosis or treatment instructions.

## Privacy checklist

- Collect only demo information needed for the prototype.
- Do not store API keys in source code.
- Clearly label mock ABHA and FHIR-style data as demo-only.

# 7. Build Plan & Expected Outcome

| | |
|---|---|
| **Phase 1** | Build dashboard and report upload |
| **Phase 2** | Add OCR and extraction |
| **Phase 3** | Add simple summaries and Telugu option |
| **Phase 4** | Save timeline and add FHIR/ABHA demo |

## Thank you

AI-Powered Personal Health Copilot • Team: _____

# Complete Feature List

Core features focused on organizing medical records and explaining health information

- Medical record upload (PDFs, images, prescriptions, lab reports)
- OCR text extraction from medical documents
- AI-powered health report summaries
- Medical terms explained in simple language
- Abnormal test values highlighted against report reference ranges
- Unified personal health profile
- Medical history and timeline management
- Health report comparison across dates
- Medicine names and prescription details extracted
- Interactive health dashboard

# Bonus & Safety Features

Additional features to improve accessibility, usefulness, and trust

- English and Telugu summaries
- Bilingual report text extraction where OCR supports it
- FHIR-style structured health records
- Mock ABHA workflow (demo only; no real account connection)
- Downloadable health summary
- Ask Your Report: questions answered from uploaded report text
- Records organized by date and category
- Privacy-conscious handling of sample/demo data
- Medical disclaimer and safety warnings
- User-friendly navigation